OBJECT ORIENTED PROGRAMMING
LAB ASSIGNMENT -1

(Topics – Basics of C++, Classes and Objects)

Programming Questions

1. Write a program (WAP) to display "Hello World" on console display. WAP to implement the following control characters:
‘\n’ is for new line, or you can use endl – cout<<endl<<“message”;
‘\t’ is for tab ; ‘\a’ is an alarm sound; ‘\r’ is carriage return to go to the beginning of the current line

```
#include <iostream>
using namespace std;
int main (){
   cout << "Hello World\n";
   cout << "Hello World\t";
   cout << "Hello World\a";
   cout << "\nHello World\r";
   return 0;
}
```

2) Write a C++ program that will ask for a temperature in Celsius and display it in degree Fahrenheit.[F=9C/5+32]

```
#include <iostream>
using namespace std;
int main(){
   int tc;
   cout << "enter the temperature in Celsius : ";
   cin>>tc;
   cout << "Temp in Fahrenheit is : " << (((9*tc)/5)+32) << endl;
   return 0;
}
```

```
~/Doc/thapar_cs_lab/oo/assignment1  master ?3  ./a.out
enter the temperature in Celsius : 25
Temp in Fahrenheit is : 77
~/Doc/thapar_cs_lab/oo/assignment1  master ?3  ./a.out
enter the temperature in Celsius : 0
Temp in Fahrenheit is : 32
~/Doc/thapar_cs_lab/oo/assignment1  master ?3  ./a.out
enter the temperature in Celsius : 100
Temp in Fahrenheit is : 212
~/Doc/thapar_cs_lab/oo/assignment1  master ?3  ./a.out
enter the temperature in Celsius : 32
Temp in Fahrenheit is : 89
```

3) WAP to demonstrate for, while, do-while (with all possible variations), like for loop can be demonstrated without giving initialization in for construct or without giving increment in for construct.

Sample:

```
for (int i=0; i<10; i++)
i=0
for (; i<10; i++)
i=0
for (; i<10;)
I++
```

```
#include <iostream>
using namespace std;
int main (){
    int n=5;
    for (int i=0;i<n;i++){
        cout << i << " ";
    }
    int i=0;
    cout << "\n";
    for (;i<n;i++){
        cout << i << " ";
    }
    cout << "\n";
    i=0;
    for (;i<n;){
        cout << i << " ";
        i++;
    }
    cout << "\n";
```

```
    i=0;
    for (;;){
        if (i==5){break;}
        cout << i << " ";
        i++;
    }
    cout << "\n";
    i=0;
    while(i<5){
        cout << i << " ";
        i++;
    }
    cout << "\n";
    i=0;
    while(true){
        if (i==5)break;
        cout << i << " ";
        i++;
    }
    cout << "\n";
    i=0;
    do{
        cout << i << " ";
        i++;
    }
    while(i<5);
    cout << "\n";
    i=10;
        do {
            cout << i <<" ";

        } while(i<5);
        cout << "\n";
    return 0;
}
```

```
~/Doc/thapar_cs_lab/oo/assignment1  master ?3  g++ q3.cpp
~/Doc/thapar_cs_lab/oo/assignment1  master ?3  ./a.out
0 1 2 3 4
0 1 2 3 4
0 1 2 3 4
0 1 2 3 4
0 1 2 3 4
0 1 2 3 4
0 1 2 3 4
10
```

4. Create a structure in C++ containing the details of Students as details below and a main function to execute the structure.
Data Members(properties):

Name
Roll No
Degree
Hostel
CurrentCGPA

Member Function(behavior):

addDetails();
updateDetails();
updateCGPA();
updateHostel();
displaydetails();

```
#include <iostream>
#include <string>
using namespace std;
struct Student{
        string Name{};
        int roll_number{};
        string Degree{};
        char hostel{};
        float cgpa{};
        void addDetails(){
            cout << "enter name : " ;
            cin >> Name;
            cout << "enter roll_number : ";
```

```
    cin>>roll_number;
    cout << "enter Degree : ";
    cin >> Degree;
    cout << "enter hostel : ";
    cin >> hostel ;
    cout << "enter cgpa : ";
    cin >> cgpa ;
}
void updateDetails(){
    cout << "press 1 to update Name\n";
    cout << "press 2 to update Roll_number\n";
    cout << "press 3 to update Degree\n";
    cout << "press 4 to update hostel\n";
    cout << "press 5 to update cgpa\n";
    cout << "press anything else  to quit\n";
    while(true){
        int x;
        cin>>x;
        switch(x){
        case 1:
            cin>>Name;
            break;
        case 2:
            cin>>roll_number;
            break;
        case 3:
            cin>>Degree;
            break;
        case 4:
            cin>>hostel;
            break;
        case 5:
            cin>>cgpa;
            break;
        default:
            return;
        }
    }
}
void updateCgpa(float cg){
    cgpa=cg;
}
void updateHostel(char host){
    hostel=host;
```

```
        }
        void displaydetails(){
            cout << "Name : " << Name << "\n";
            cout << "Roll Number : " << roll_number << "\n";
            cout << "Degree : " << Degree << "\n";
            cout << "Hostel : " << hostel << "\n";
            cout << "cgpa : " << cgpa << "\n";
        }
};
int main (){
    Student s1;
    s1.addDetails();
    s1.updateCgpa(7.8);
    s1.updateHostel('A');
    s1.displaydetails();
    s1.updateDetails();
    s1.displaydetails();
   return 0;
}
```

```
enter name : ud
enter roll_number : 94
enter Degree : BTECH
enter hostel : J
enter cgpa : 8.2
Name : ud
Roll Number : 94
Degree : BTECH
Hostel : A
cgpa : 7.8
press 1 to update Name
press 2 to update Roll_number
press 3 to update Degree
press 4 to update hostel
press 5 to update cgpa
press anything else  to quit
4
M
5
7.2
9
Name : ud
Roll Number : 94
Degree : BTECH
Hostel : M
cgpa : 7.2
```

q5) Differentiate between private and public access/scope. Perform the question no. 4 with class instead of structure with having the data members private and some member functions in private scope and some in public scope.

```
#include <iostream>
#include <string>
using namespace std;
class Student{
   private:
      string Name{};
      int roll_number{};
      string Degree{};
      char hostel{};
      float cgpa{};
   public:
      void addDetails(){
         cout << "enter name : " ;
         cin >> Name;
         cout << "enter roll_number : ";
         cin>>roll_number;
         cout << "enter Degree : ";
         cin >> Degree;
         cout << "enter hostel : ";
         cin >> hostel ;
         cout << "enter cgpa : ";
         cin >> cgpa ;
      }
      void updateDetails(){
         cout << "press 1 to update Name\n";
         cout << "press 2 to update Roll_number\n";
         cout << "press 3 to update Degree\n";
         cout << "press 4 to update hostel\n";
         cout << "press 5 to update cgpa\n";
         cout << "press anything else  to quit\n";
         while(true){
            int x;
            cin>>x;
            switch(x){
            case 1:
               cin>>Name;
               break;
            case 2:
               cin>>roll_number;
               break;
```

```
            case 3:
                cin>>Degree;
                break;
            case 4:
                cin>>hostel;
                break;
            case 5:
                cin>>cgpa;
                break;
            default:
                return;
            }
        }
    }
    void updateCgpa(float cg){
        cgpa=cg;
    }
    void updateHostel(char host){
        hostel=host;
    }
    void displaydetails(){
        cout << "Name : " << Name << "\n";
        cout << "Roll Number : " << roll_number << "\n";
        cout << "Degree : " << Degree << "\n";
        cout << "Hostel : " << hostel << "\n";
        cout << "cgpa : " << cgpa << "\n";
    }
};
int main (){
    Student s1;
    s1.addDetails();
    s1.updateCgpa(7.8);
    s1.updateHostel('A');
    s1.displaydetails();
    s1.updateDetails();
    s1.displaydetails();
   return 0;
}
```

```
enter name : udbhav
enter roll_number : 121
enter Degree : BE
enter hostel : A
enter cgpa : 8.2
Name : udbhav
Roll Number : 121
Degree : BE
Hostel : A
cgpa : 7.8
press 1 to update Name
press 2 to update Roll_number
press 3 to update Degree
press 4 to update hostel
press 5 to update cgpa
press anything else  to quit
2
94
4
B
9
Name : udbhav
Roll Number : 94
Degree : BE
Hostel : B
cgpa : 7.8
```

q6. Create a code snippet that illustrates the following:
a. Calling of private member functions inside public member function
b. Access private member functions inside public member function

```
#include <iostream>
using namespace std;
class A{
   private:
      void privatefucntion(){
         cout << "this is a private function\n";
      }
   public:
      void callsprivate(){
         cout << "this function calls private member  ";
         privatefucntion();
         void (A::*ptr)()= &A::privatefucntion;
         (this->*ptr)();
      }
};
int main (){
   A aa;
   aa.callsprivate();
   return 0;
}
```

```
this function calls private member  this is a private function
this is a private function
```

q7) Define a class named Complex with properties (real and imaginary) and methods as per following details.
void set () to initialize object values.
void display () to display complex number.
Complex sum (Complex) or void sum (Complex) to add two complex numbers (objects of Complex class) and return complex_number (object of Complex class) as result. Properties (real and imaginary) of the code should have private access modifier and member functions should have public access modifier in C++ class.

```
#include <iostream>
using namespace std;
class Complex{
```

```
    private:
        float real;
        float imag;
    public:
        void set(){
            cout << "enter real value : ";
            cin>>real;
            cout << "enter imag value : ";
            cin>>imag;
        }
        void display(){
            cout << real << "+ i" << imag << endl;
        }
        void setreal(float r){
            real=r;
        }
        void setimag(float i){
            imag=i;
        }
        float getreal(){
            return real;
        }
        float getimag(){
            return imag;
        }
};
        Complex sum(Complex *a , Complex *b){
            Complex result;
            float tr=a->getreal()+b->getreal();
            result.setreal(tr);
            float ti=a->getimag()+b->getimag();
            result.setimag(ti);
            return result;
        }
int main (){
    Complex a;
    Complex b;
    a.set();
    cout << "complex a-> ";
    a.display();
    b.set();
    cout << "complex b-> ";
    b.display();
    Complex c=sum(&a,&b);
```

```
    cout << "their sum -> ";
    c.display();
    return 0;
}
```

```
~/Doc/thapar_cs_lab/oo/assignment1   master ?3   ./a.out
enter real value : 10
enter imag value : 5
complex a-> 10+ i5
enter real value : 92
enter imag value : -2
complex b-> 92+ i-2
their sum -> 102+ i3
~/Doc/thapar_cs_lab/oo/assignment1   master ?3   ./a.out
enter real value : -1
enter imag value : -2
complex a-> -1+ i-2
enter real value : 3
enter imag value : 2
complex b-> 3+ i2
their sum -> 2+ i0
```

q8) Implement namespace in a program to illustrate the use of same name variables and functions in different sections/libraries of the code.

```
#include <iostream>
using namespace std;

namespace A {
    int x = 10;
    void show() {
        cout << "A x = " << x << endl;
    }
}

namespace B {
    int x = 20;
    void show() {
        cout << "B x = " << x << endl;
    }
}
```

```
int main() {
    A::show();
    B::show();
    return 0;
}
```

```
~/Doc/thapar_cs_lab/oo/assignment1  master ?3  ./a.out
A x = 10
B x = 20
```